

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2023-24

Paper: BVCA-405 — Android Programming

Time: Three Hours

Maximum Marks: 70

Instructions: Answer the questions in each section as instructed. All questions carry marks as indicated.

SECTION — A

Long-Answer Questions. Answer any two questions:

2 × 12 = 24 Marks

Question 1. Discuss the Android Architecture. Explain the importance and functions of the different layers in detail.

Question 2. Discuss the directory structure of an Android project in detail.

Question 3. What are the features of the Android Operating System? Discuss the advantages and disadvantages of Android. Also write about its major version releases and breakthroughs.

Question 4. What is ART (Android Run Time)? Discuss the features and working of ART.

SECTION — B

Question 5. Attempt any six parts (Answer in about 200 words each): **6 × 6 = 36 Marks**

- (a) What is the `AndroidManifest.xml` file? Write down its purpose and list its key components.
- (b) What is an APK file? Discuss in detail how you can build an APK file in Flutter.
- (c) What is the difference between `ListView` and `ListView.builder` widgets in Flutter? Explain with examples.
- (d) What are the key differences between Dalvik Virtual Machine (DVM) and Android Runtime (ART)?
- (e) Explain:
 - (i) What are the different types of widgets in Flutter?
 - (ii) What is the purpose of the `Scaffold` widget in Flutter?
- (f) Name three common image file formats used in mobile applications. Explain the characteristics and appropriate usage scenarios for each.
- (g) What is the difference between a programming language and a framework? How is Flutter different from native development languages like Kotlin (for Android) and Swift (for iOS)?

- (h) Discuss the architecture of the Flutter framework. Explain the inner working of its key components.
- (i) Briefly explain the purpose and usage of the following in Flutter:
 - (i) `MaterialApp` widget
 - (ii) `setState()` method

SECTION — C

Question 6. Attempt all parts:

10 × 1 = 10 Marks

- (a) Which file format is commonly used for storing persistent data in Android applications?
- (b) What is the primary advantage of using Flutter for cross-platform app development?
- (c) Which widget is used to render elements side by side in Flutter?
- (d) Which programming language is primarily used for developing native Android applications?
- (e) Which database engine is most commonly embedded and used in mobile applications?
- (f) Which widget is commonly used to add margins/padding around widgets in Flutter?
- (g) How can you retrieve the screen dimensions of a device in Flutter? Write the name of the utility class used.
- (h) What is the API level of the latest stable Android version?
- (i) Which programming language is used in the Flutter framework?
- (j) Which file in an Android project contains the required application permissions?